DSLR: Deep Stacked Laplacian Restorer for Low-Light Image Enhancement

Seokjae Lim and Wonjun Kim, *Member, IEEE*

***Abstract*—Various images captured in complicated lighting conditions often suffer from deterioration of the image quality. Such poor quality not only dissatisfies the user expectation but also may lead to a significant performance drop in many applications. In this paper, anovel method for low-light image enhancement is proposed by leveraging useful propertiesof the Laplacian pyramid both in image and feature spaces. Specifically, the proposed method, so-called a deep stacked Laplacian restorer (DSLR), is capable of separately recovering the global illumination and local details from the original input, and progressively combining them in the image space. Moreover, the Laplacian pyramid defined in the feature space makes such recovering processes more efficient based on abundant connectionsof higher-order residuals in a multiscale structure. This decomposition-based scheme is fairly desirable for learning the highly nonlinear relation between degraded images and their enhanced results. Experimental results on various datasets demonstrate that the proposed DSLR outperforms state-of-the-art methods. The code and model are publicly available at: https://github.com/SeokjaeLIM/DSLR-release.**

***Index Terms*—Low-light image enhancement, Laplacian pyramid, deep-stacked laplacian restorer (DSLR), decomposition-based scheme.**

I. Introduction

As CAMERAS on various mobile devices become readily-available and more wide-spread, many people have recently begun to show an interest in taking pictures in their daily life. Although technologies related to the camera sensor have been steadily developed for user satisfaction on the image quality, images acquired from real-world environments are often degraded due to complicated lighting conditions, e.g., night time, uneven illumination, back-lighting, etc. Such images probably contain degradation of the visual quality and thus adversely affect the performance in many applications, such as object detection [1] and tracking [2]. As an example, some failure cases of object detection caused by various lighting conditions are shown in Fig. 1.



Fig. 1. Several examples of object detection [1]. Top: detection results on degraded images. Bottom: detection results on images restored by the proposed method. Note that missing parts when using degraded images are successfully detected in enhancement results by the proposed method.

To make the buried information visible, various methods have been proposed for last few decades. In the beginning, many researchers attempted to directly flatten the distribution of pixel intensities or colors for stretching the dynamic range of the input image based on statistical information. Even though these methods, e.g., histogram equalization (HE) and its variants [3], [4], [5], are quite simple and effective for contrast enhancement, they often fail to moderately recover the contrast (i.e., under- or over-enhancement) in the input image due to limitation of the probabilistic analysis without considering spatial associations. On the other hand, several approaches have actively employed the Retinex theory [6] that images can be decomposed into illumination and reflectance layers. Most Retinex-based methods [7], [8] firstly estimate the illumination component from the given image and separate the reflectance one, which is treated as a well-exposed image in the early studies [9], [10]. However, the ambiguousness of the decomposition result often yields unnatural outputs, which give the visual discomfort to users. Most recently, owing to the plentiful possibilities of the generative model via the deep neural network (DNN), the problem of low-light image enhancement has been reformulated as the problem of image-to-image translation, that is, enhancement results can be now restored without any physical assumption. Even though various DNN-based models [11], [12], [13] greatly improve the visual quality of restored results, they still have respective limitations on learning the highly nonlinear process of the photographic adjustment.

In this paper, a deep stacked Laplacian restorer (DSLR) is proposed for low-light image enhancement. The key idea of the proposed method is to adjust the global illumination and restore local details by exploiting a Laplacian pyramid [14] both in image and feature spaces. Specifically, such two tasks are separately

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The authors are with the Department of Electrical and Electronics Engineering, Konkuk University, Seoul 05029, South Korea (e-mail: hgg08@konkuk.ac.kr; wonjkim@konkuk.ac.kr).

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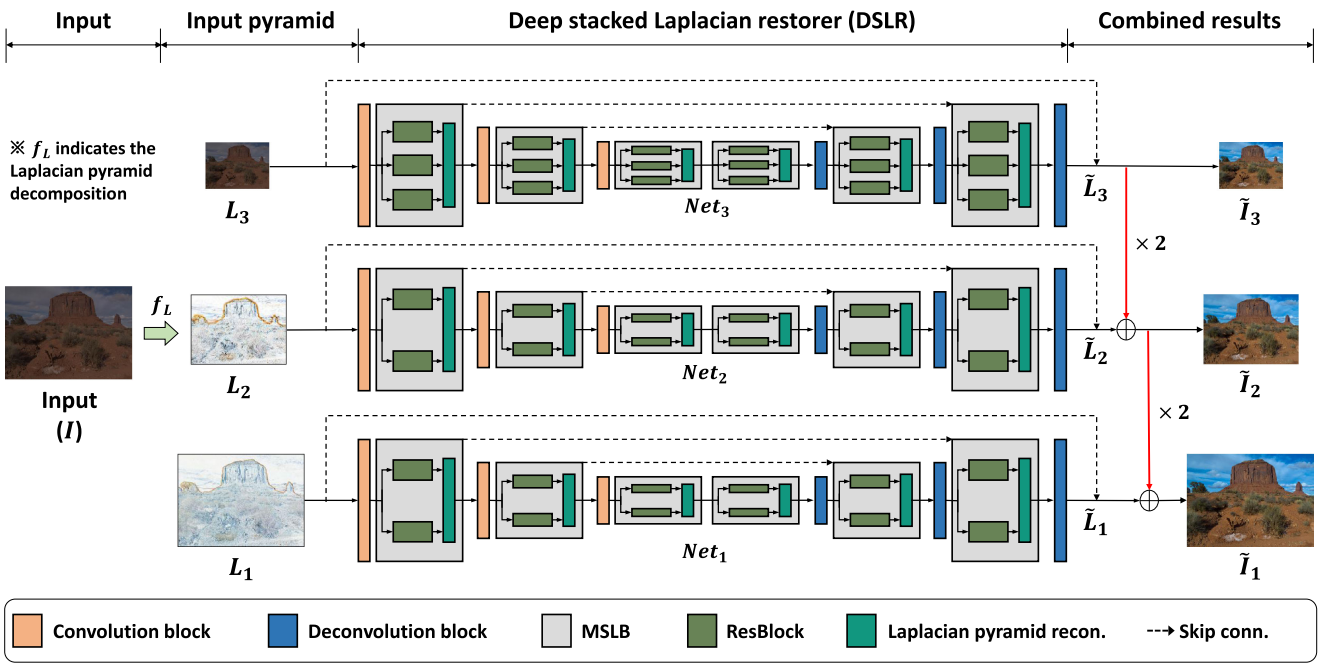


Fig. 2. Overall architecture of the proposed method for low-light image enhancement. The decomposition technique is firstly applied to the input image for constructing a three-level Laplacian pyramid. Such decomposed images are then restored by corresponding networks at each pyramid level. Finally, restored images are combined via up-sampling followed by pixel-wise addition for generating the final enhancement result (i.e., $\tilde{I}_1$). Note that pixel values of inputs at finer pyramid levels (i.e., L_1 and L_2) are inverted in this figure for better visualization.

trained in different streams of the encoder-decoder architecture, which are defined by decomposition of the Laplacian pyramid as shown in Fig. 2, and corresponding outputs are progressively combined to reconstruct the enhancement result of the input image. Moreover, the Laplacian pyramid applied to the feature space makes optimization easier based on abundant connections of higher-order residuals in a multiscale structure, which lead to a stable convergence with the relatively small loss during the training phase. In contrast to the simple encoder-decoder architectures, which have been widely employed in previous methods, one important advantage of the proposed DSLR is that it has an ability to effectively preserve local details without significant color distortions by using such connections of residuals learned in the multi-level Laplacian pyramid. The main contributions of the proposed method can be summarized as follows:

1) We propose a Laplacian pyramid based multiscale network, called a deep stacked Laplacian restorer (DSLR), for resolving the problem of low-light image enhancement.
2) To make the learning process efficient, we define a multiscale Laplacian-residual block (MSLB). This block plays an important role to improve the flow of information based on abundant connections of higher-order residuals, which are defined in a multiscale structure of the embedding feature space, during the training phase. This makes the proposed method yield more reliable results for low-light image enhancement.
3) We perform quantitative and qualitative evaluations on the MIT-Adobe FiveK dataset [15] and our own dataset, which is constructed based on samples from Google image search. Experimental results demonstrate that the proposed DSLR significantly improves the performance of low-light image enhancement (over 1.24 dB in an aspect of PSNR) compared to state-of-the-art methods.

The remainder of this paper is organized as follows. A comparative review of related works is presented in Section II. The proposed DSLR is explained in detail in Section III. Experimental results are demonstrated with various datasets in Section IV. The conclusions follow in Section V.

II. Related Work

In this Section, we give a brief review of studies related to the Laplacian pyramid and image enhancement, which can be divided into traditional (i.e., handcrafted feature-based) methods and learning-based methods.

1) Laplacian Pyramid: The Laplacian pyramid [14] was first designed for compact image representation with simple operations, i.e., reducing and expanding. Many researchers have attempted to apply the Laplacian pyramid to a wide range of computer vision tasks, e.g., texture analysis and synthesis [16], edge-aware filtering [17], etc. More recently, some researchers have focused on the property that the Laplacian pyramid has a linearly invertible process to reconstruct the original input without any loss of information for refining local details accurately. For example, Ghiasi *et al.* [18] proposed the refinement process based on the Laplacian pyramid to enhance local details of the estimated result in the low resolution for semantic segmentation. Authors of [19] progressively restored the degraded resolution of the input by predicting the high-frequency

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residuals, i.e., residuals of finer pyramid levels, from latent features of the coarse resolution. Inspired by those researches, we proposed a deep stacked Laplacian restorer (DSLR) for low-light image enhancement. The proposed method differs from previous approaches in two aspects. First, we leveraged useful properties of the Laplacian pyramid in the feature space. Specifically, the number of skip paths is efficiently increased by adopting the Laplacian pyramid for residual learning in the feature space. This is helpful to improve the flow of information during the training phase, which leads to the stable convergence with the relatively small loss. Moreover, unlike previous methods, which only focused on the reconstruction process of the Laplacian pyramid [18], [19], we also proposed to utilize both decomposition and reconstruction processes of the Laplacian pyramid in the image space for separately restoring the global illumination and local details of underexposed images.

2) Traditional Methods: Histogram equalization [3] and its variants [20], [21] have been popularly adopted for image enhancement. Among them, the local equalization scheme with the clipping constraint, so-called CLAHE [22], has shown the remarkable performance without severe distortions of the semantic information for a given scene. On the other hand, various optimization techniques have been employed to adaptively stretch the distribution of intensities or colors by forcing regularization effects to the histogram, e.g., minimizing the weighted distance between counts of histograms [23] and the dissimilarity of diagonal elements between two-dimensional histograms [24]. The relation of neighboring pixels also has been actively utilized to take into account spatial associations, which give a great help to find the optimal histogram mapping for the automatic contrast change [25]. More recently, many studies have started to focus on decomposing the scene intensity into illumination and reflectance components based on the Retinex theory [6]. Most approaches belonging to this category firstly adjust the dynamic range of the estimated illumination and subsequently re-combine the corresponding result with the separated reflectance for generating the enhancement result. In particular, Wang *et al.* [26] proposed a bright-pass filter, which restricts the range of the reflectance component in 0 to 1, to estimate illumination and reflectance components. Guo *et al.* [27] employed the maximum value of RGB channels at each pixel position as the initial illumination map and refined it based on a structure prior via an augmented Lagrangian multiplier based algorithm. Furthermore, several researchers attempted to simultaneously extract illumination and reflectance components based on the weighted variational model [8]. However, their decomposition procedures do not consider the noise problem, thus additional denoising techniques are needed to suppress unnecessary noises in the enhancement result. To alleviate this problem, Li *et al.* [28] designed their objective function with the additional noise term and resolved it via the optimization technique. Kim *et al.* [29] tried to estimate the illumination component by adopting the maximal value in multiple diffusion spaces at each pixel position, so-called maximal diffusion value, and showed the reliable enhancement results without the problem of noise amplification. Even though traditional methods improve the visual quality of degraded images quite well, those primarily focus on the improvement of the contrast and thus often fail to maintain the naturalness of color attributes under various real-world environments.

3) Learning-Based Methods: Owing to the great success of the deep neural network (DNN), the problem of image enhancement can now be resolved by exploiting the DNN-based generative model. In the beginning stage, a simple encoder-decoder architecture, which is based on stacked convolution layers, has been adopted with various loss functions to accurately restore the enhanced result. For example, the consistency of color attributes, textural complexities, and the type of contents have been popularly considered for loss functions [30] while the guiding factor also has been employed to suppress noise artifacts occurring during the enhancement process [31]. Most recently, more complicated architectures have been developed to efficiently allow for the highly nonlinear process of image enhancement. Instead of directly learning the complicated relation between inputs and enhancement results, the architecture for the image-to-illumination translation was introduced with the bilateral grid-based upsampling technique [13]. Xu *et al.* [32] proposed to decompose the input image into high and low frequency components, and adaptively enhance them with attention to context encoding (ACE) and cross domain transformation (CDT) modules. On the other hand, many unpaired learning schemes have been developed due to the lack of the paired training data. For example, Chen *et al.* [12] proposed the two-way generative adversarial network (GAN), which significantly improves the visual quality of enhancement results. Similarly, Jiang *et al.* [33] proposed GAN based network with dual discriminators to balance global and local image enhancement, and trained such network via a self-regularized perceptual loss. Furthermore, several researchers proposed the zero-reference deep curve estimation, which estimates the pixel-wise and high-order curves for dynamic range adjustment of a given image, considering the pixel value range, monotonicity and differentiability with non-reference loss functions [34].

In this paper, we propose a deep stacked Laplacian restorer (DSLR) for low-light image enhancement. Rather than using only a single stream of the enhancement process, the proposed method focuses on the decomposition strategy in a multiscale framework. Technical details of the proposed method will be explained in the following Section.

III. Proposed Method

The heart of the proposed method, i.e., deep stacked Laplacian restorer (DSLR), lies in applying the Laplacian pyramid to both image and feature spaces. Specifically, Laplacian residuals at finer pyramid levels guide encoder-decoder architectures to accurately restore local details while the coarsest level of Laplacian pyramids forces the relevant network to adjust the global illumination as shown in Fig. 2. In each stream of the network, features are also decomposed into Laplacian pyramids by using our proposed multiscale Laplacian-residual block (MSLB). This gives a great help to resolve gradient-vanishing issues by utilizing abundant connections of higher-order residuals in a multiscale manner. By incorporating the Laplacian pyramid both into

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image and feature spaces, natural colors as well as the contrast are restored more accurately from degraded images.

A. Architecture Details

The overall architecture of the proposed method is shown in Fig. 2. The proposed DSLR basically consists of three encoder-decoder streams, one of which is in charge of learning relevant features for low-light image enhancement at the corresponding scale defined by decomposition of the Laplacian pyramid. Each restorer, i.e., a single encoder-decoder architecture, can be divided into three main components, which are convolution blocks, deconvolution blocks, and our proposed MSLBs. At the encoder side, features are efficiently extracted by using abundant connections of higher-order residuals, which are the key trait of the proposed MSLB. Note that details of MSLB will be introduced at the latter part of this subsection.

Based on the symmetric property of the encoder-decoder architecture, latent features extracted by the encoder are transformed back to the shape of the original input through the decoder. After this decoding process at each pyramid level, the restored result is up-sampled and pixel-wisely added to the output estimated from the finer level of the Laplacian pyramid. For all convolution layers, the filter size is set to 3×3 in MSLBs and the size of 5×5 is used otherwise. We set the filter size to be 4×4 for deconvolution layers. Note that instance normalization [35] and ReLU layers follow by all the convolution and deconvolution layers except the final output layer. The architecture details are shown in Table I.

1) Laplacian Pyramid in the Image Space: The input image is firstly decomposed into the three-level Laplacian pyramid, which can be formulated as follows:

$$L_k = I_k - f_{\uparrow}(I_{k+1}), \quad k = 1, 2, \tag{1}$$

where k denotes the level index in the Laplacian pyramid. I_k and L_k are the resized version of the original input to $1/2^{k-1}$ scale and the Laplacian residual image computed at the k-th pyramid level, respectively. $f_{\uparrow}(\cdot)$ denotes the function of up-sampling ($\times 2$). For all resizing procedures in our method, we employ the bilinear interpolation. Note that L_3 is set equal to I_3, which is acquired by down-sampling the original input to the 1/4 scale (i.e., the coarsest level). To seek optimal parameters describing the relation between the degraded image and the enhancement result in a supervised manner, the image retouched by the expert, i.e., ground truth, is also decomposed in the same way.

Based on this, we build a set of restorers, i.e., Net_1, Net_2, and Net_3, to separately learn the enhancement process by using Laplacian residual images. Specifically, the restorer at the coarsest level, i.e., Net_3, adjusts the global illumination and others enhance underlying details at finer pyramid levels. The output of Net_3 is up-sampled and pixel-wisely added to the Laplacian residual, which is restored by Net_2, to generate the intermediate result. The final result is obtained by up-sampling this intermediate result and adding it to the restored residual via Net_1. The overall process is expressed as follows:

$$\tilde{L}_k = Net_k(L_k), \quad k = 1, 2, 3, \tag{2}$$

TABLE I
THE DETAILED ARCHITECTURE OF EACH RESTORER FOR THE PROPOSED DSLR

Module	Layer type	Weight dimension	Stride
	Net 1 and Net 2 (restorers at finer levels)		
Encoder	Conv	32 × 3 × 5 × 5	2
	4× Conv (MSLB, W = 2)	32 × 32 × 3 × 3	1
	Conv	64 × 32 × 5 × 5	2
	4× Conv (MSLB, W = 2)	64 × 64 × 3 × 3	1
	Conv	128 × 64 × 5 × 5	2
	4× Conv (MSLB, W = 2)	128 × 128 × 3 × 3	1
Decoder	4× Conv (MSLB, W = 2)	128 × 128 × 3 × 3	1
	DeConv	64 × 128 × 4 × 4	2
	4× Conv (MSLB, W = 2)	64 × 64 × 3 × 3	1
	DeConv	32 × 64 × 4 × 4	2
	4× Conv (MSLB, W = 2)	32 × 32 × 3 × 3	1
	DeConv	3 × 32 × 4 × 4	2
	Net 3 (restorer at the coarsest level)		
Module	Layer type	Weight dimension	Stride
Encoder	Conv	64 × 3 × 5 × 5	2
	6× Conv (MSLB, W = 3)	64 × 64 × 3 × 3	1
	Conv	128 × 64 × 5 × 5	2
	6× Conv (MSLB, W = 3)	128 × 128 × 3 × 3	1
	Conv	256 × 128 × 5 × 5	2
	6× Conv (MSLB, W = 3)	256 × 256 × 3 × 3	1
Decoder	6× Conv (MSLB, W = 3)	256 × 256 × 3 × 3	1
	DeConv	128 × 256 × 4 × 4	2
	6× Conv (MSLB, W = 3)	128 × 128 × 3 × 3	1
	DeConv	64 × 128 × 4 × 4	2
	6× Conv (MSLB, W = 3)	64 × 64 × 3 × 3	1
	DeConv	3 × 64 × 4 × 4	2

Note: the weight dimension is formulated as the number of output channels × the number of input channels × the height of the filter × the width of the filter. W indicates the number of ResBlocks for constructing the proposed MSLB. The padding size is set to 1 through all the layers.

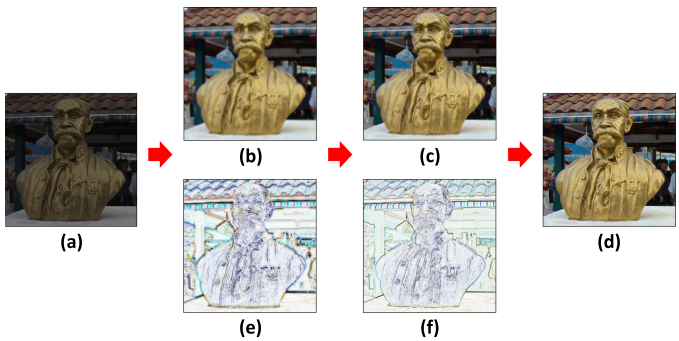


Fig. 3. (a) Input image. (b) $\tilde{I}_3$: the enhancement result at the coarsest pyramid level. (c) $\tilde{I}_2$: the enhancement result at the mid pyramid level, i.e., (b) + (e). (d) $\tilde{I}_1$: the final enhancement result of our DSLR, i.e., (c) + (f). (e) $\tilde{L}_2$: the restored residual at the mid pyramid level. (f) $\tilde{L}_1$: the restored residual at the finest pyramid level. Note that we invert restored residuals at finer pyramid levels and resize all images to the input scale for better visualization.

$$\tilde{I}_k = \tilde{L}_k + f_{\uparrow}(\tilde{I}_{k+1}), \quad k = 1, 2, \tag{3}$$

where $\tilde{L}_k$ is the restored residual through the encoder-decoder architecture at the k-th pyramid level, i.e., Net_k. The final output, i.e., $\tilde{I}_1$, is generated by progressively combining $\tilde{L}_k$ across pyramid levels via (3). Note that $\tilde{I}_3$ is the same as $\tilde{L}_3$. The corresponding results are shown in Fig. 3.

2) Laplacian Pyramid in the Feature Space: Inspired by nested skip connections [36], we propose a multiscale Laplacian-residual block (MSLB) based on abundant connections of higher-order residuals in a multiscale framework. First of all, the

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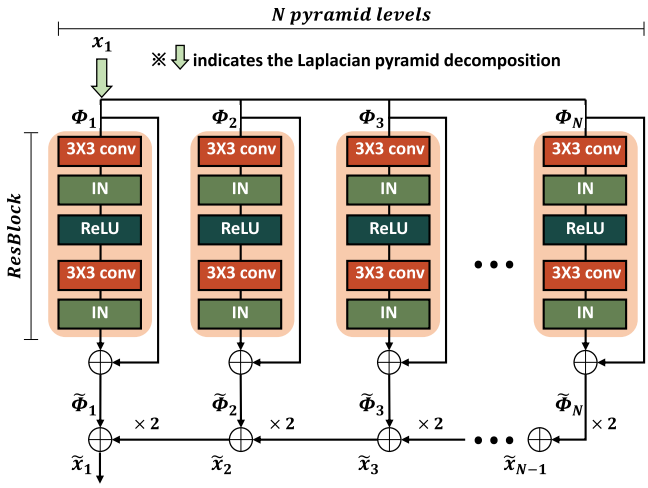


Fig. 4. Proposed multiscale Laplacian-residual block (MSLB) constructed by N Laplacian pyramid levels. Note that our MSLB in a single pyramid level performs the same operation as a single ResBlock.

latent feature acquired from preceding convolution and deconvolution blocks is decomposed into a sequence of N Laplacian residuals, i.e., $\Phi_1, \Phi_2, \ldots, \Phi_N$. Thus, for $1 \leq k < N$,

$$\Phi_k = x_k - f_{\uparrow}(x_{k+1}), \tag{4}$$

where x_k denotes the resized feature, which is obtained by downsampling the input feature to the $1/2^{k-1}$ scale. In the case of the coarsest pyramid level (i.e., $k = N$), Φ_N is given as x_N which represents the resized input by using the bilinear interpolation to fit the coarsest level (i.e., $1/2^{N-1}$ scale). The decomposed feature, i.e., Φ_k, is then fed into the single ResBlock as follows:

$$\tilde{\Phi}_k = \Phi_k + F_k(\Phi_k), \quad k = 1, 2, 3, \ldots, N, \tag{5}$$

where F_k represents the function of the single ResBlock in the k-th pyramid level. Note that we slightly modify the original version of ResBlock [37] by replacing the batch normalization [38] with the instance normalization [35]. Finally, the input feature is refined based on the following reconstruction process, given as

$$\tilde{x}_k = \tilde{\Phi}_k + f_{\uparrow}(\tilde{x}_{k+1}), \quad k = 1, 2, 3, \ldots, N-1. \tag{6}$$

Note that this process starts with $\tilde{x}_N = \tilde{\Phi}_N$ (i.e., the coarsest level). In contrast to N skip connections used in conventionally stacked N ResBlocks, the number of skip paths is increased as $(2N-1)$ residual connections, which are calculated by adding N connections of all ResBlocks in our MSLB to $(N-1)$ connections required for the procedure of Laplacian pyramid reconstruction in (6). The detailed architecture of MSLB is shown in Fig. 4. As similar to [36] which investigated the capability of the residual connection, one important advantage of the proposed MSLB is that difficulties in interpreting the complicated process of low-light image enhancement can be efficiently alleviated by utilizing such abundant connections in a multiscale framework, and the flow of information is also remarkably improved in the training phase. Note that Net_3 of our DSLR serves to transform the input into a well-exposed output by tuning the global lighting condition, therefore the three-level MSLB (i.e., $N = 3$) is adopted to comprehend the overall trend of degradation in the input image via the wider receptive field. On the other hand, the two-level MSLB (i.e., $N = 2$) is employed to restore Laplacian residual images in Net_1 and Net_2 since such images can be restored with the relatively small receptive field due to their sparse properties as shown in Fig. 3(e) and (f). Some examples of the visualized feature maps produced by MSLBs are shown in Fig. 5.

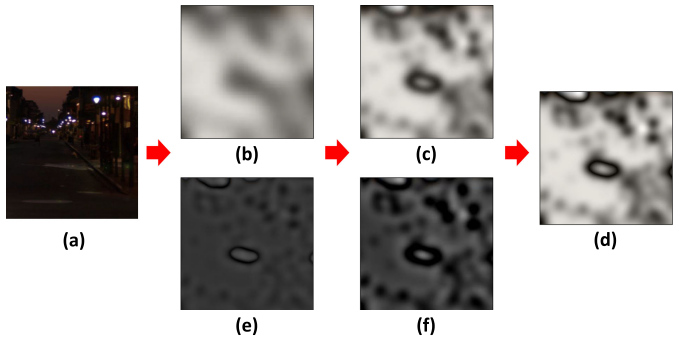


Fig. 5. Examples of the visualized feature maps produced by MSLBs. (a) Test image. (b) $\tilde{x}_3$. (c) $\tilde{x}_2$. (d) $\tilde{x}_1$. (e) $\tilde{\Phi}_2$. (f) $\tilde{\Phi}_1$.

B. Loss Function

To optimize trainable parameters in the proposed DSLR, we formulate a loss function $\mathcal{L}_{total}$ which is composed of three components, i.e., data loss $\mathcal{L}_d$, Laplacian loss $\mathcal{L}_l$, and color loss $\mathcal{L}_c$ as follows:

$$\mathcal{L}_{total} = \omega_d \mathcal{L}_d + \omega_l \mathcal{L}_l + \omega_c \mathcal{L}_c, \tag{7}$$

where ω_d, ω_l, and ω_c denote the weight value for each loss term, respectively. Based on extensive experiments, those are set to 2, 2, and 1 for the best performance.

1) Data Loss: The mean squared error, i.e., ℓ_2 loss, is one of the most well-known loss functions in the field of optimization since it is convex and differentiable. This ℓ_2 loss can guide the enhancement result to be similar with the corresponding ground truth in a viewpoint of the pixel-wise average, and we thus employ it as our data loss, which is defined as follows:

$$\mathcal{L}_d = \sum_{k=1}^{3} \frac{1}{T_k} ||\tilde{I}_k - I_k^*||^2, \tag{8}$$

where $\tilde{I}_k$ and I_k^* are the enhancement result in the k-th Laplacian pyramid level obtained by using (3) and the corresponding ground truth, respectively. The loss computed at each pyramid level is normalized by T_k, which is the total number of pixels contained in $\tilde{I}_k$.

2) Laplacian Loss: Despite useful properties of ℓ_2 loss, it often yields blurry restoration results due to the squared term, which gives low gradients for relatively small errors as described in [39]. To preserve the local sharpness of the enhancement result, we define the Laplacian loss by using ℓ_1 loss as follows:

$$\mathcal{L}_l = \sum_{k=1}^{2} \frac{1}{T_k} |\tilde{L}_k - L_k^*|, \tag{9}$$

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Fig. 6. Visual comparison of the proposed method according to different combinations of loss functions on the MIT-Adobe FiveK dataset.

where $\tilde{L}_k$ and L_k^* denote the Laplacian residual image restored via (2) and its ground truth in the k-th Laplacian pyramid level, respectively. For the Laplacian loss, T_k is set to the total number of pixels in $\tilde{L}_k$. The Laplacian loss is suitable to train the restoration process of the sparse image containing a lot of zero values (e.g., inputs of finer pyramid levels, i.e., $k = 1$ and 2, in the proposed method) since it propagates constant gradients regardless of the magnitude of the error.

3) Color Loss: To ensure that the color vectors, which are composed of R, G, and B channels of the predicted enhancement result, have the same direction with those of corresponding ground truth, we slightly modify the color loss introduced in [13] by utilizing the cosine similarity [40] as follows:

$$\mathcal{L}_c = \sum_{k=1}^{3} \left(1 - \frac{1}{H_k \times W_k} \sum_{i \in (H_k, W_k)} \frac{\tilde{I}_k(i) \cdot I_k^*(i)}{||\tilde{I}_k(i)||^2 ||I_k^*(i)||^2} \right), \tag{10}$$

where H_k and W_k are the height and the width of the enhancement result in the k-th Laplacian pyramid level, respectively. i denotes the pixel position of $\tilde{I}_k$. $\cdot$ indicates the inner product. The closer the color is, the closer the color loss $\mathcal{L}_c$ is to zero. By considering directions of color vectors in the restoration process, color attributes of enhancement results are successfully restored as similar to the ground truth.

The ablation study which demonstrates the effectiveness of each loss function is shown in Fig. 6. The local details in the enhancement result are sharply reconstructed by adopting the Laplacian loss $\mathcal{L}_l$ (see the third image in Fig. 6) while the colors, which seem to be washed away, in the third image are re-generated as vivid colors by using the color loss $\mathcal{L}_c$ as shown in the fourth image of Fig. 6.

IV. Experimental Results

In this Section, we demonstrate the superiority of the proposed method through various experimental results on the MIT-Adobe FiveK dataset [15], which is widely employed for supervised learning-based low-light image enhancement, and our own dataset constructed based on Google image search. To be specific, we firstly explain the training procedure of the proposed method. The performance comparisons with state-of-the-art approaches are explained in detail and the ablation study finally follows.

A. Training

The proposed method is trained based on the MIT-Adobe FiveK dataset, which contains 5,000 underexposed images and corresponding five enhanced versions produced by different experts (A/B/C/D/E). We only use the output retouched by the expert C, which is employed as the ground truth in most previous methods [12], [13], [41], [42], for the supervised learning. Specifically, 4,500 pairs are randomly selected for training and the performance of our proposed method is evaluated on remaining samples in the same way introduced in [13]. To avoid the overfitting problem, training samples are randomly cropped with the resolution of 1024×1024 pixels, followed by randomly flipping them with respect to horizontal and vertical axes.

The proposed method is implemented on the PyTorch framework [43]. The Adam optimizer [44] is used to train the proposed DSLR for 300 epochs with a batch size of 5 where power and momentum are set to 0.9 and 0.999, respectively. The initial learning rate is set to 3×10^{-4} and reduced by 1.2% of the current value at every epoch. It takes 60 hours for training the proposed DSLR using two NVIDIA GeForce Titan XP GPUs.

B. Performance Evaluations

To demonstrate the efficiency and robustness, the proposed method is compared with three traditional methods, which are known as CLAHE [22], LDR [25], and LIME [27], and five learning-based methods, i.e., HDRNet [11], DR [42], SICE [45], DPE [12], and UPE [13]. We produce their results by using the publicly available source codes for the fair comparison. Specifically, four learning-based methods [11]–[13], [42] are re-trained with the parameter setting recommended by authors on the MIT-Adobe FiveK dataset since our composition of training and test samples is slightly different from that of their methods. Furthermore, we collect 100 natural images from Google image search with keywords, e.g., underexposed, low-light, night-time, and casting shadows, to verify the generality of the proposed DSLR. More concretely, the constructed dataset covers a wide range of landscapes, e.g., city, forest, beach, river, indoor environment, etc. The average pixel value of the constructed dataset is about 79, and the size of images in our dataset varies from 1920×1080 pixels to 8000×6000 pixels. Note that all the experiments for testing each method were performed on a PC with an Intel i7-6850K@3.60 GHz CPU and an NVIDIA GeForce Titan XP GPU.

First of all, some examples of enhancement results on the MIT-Adobe FiveK dataset are shown in Figs. 7 and 8. More concretely, three traditional methods, i.e., CLAHE, LDR, and LIME, are able to improve the visibility of the original input, but they still fail to restore heavily degraded colors to be vivid. In the cases of four learning-based methods [11]–[13], [42], they show somewhat colorful results. However, colors in enhancement results of HDRNet and UPE relatively seem to be washed away compared to those of the proposed method. DR and DPE have a limitation in moderately adjusting the overall brightness of enhancement results. In contrast to previous approaches, our DSLR successfully restores the input image with vivid colors like the ground truth (see colors of the maple tree and the squirrel shown in Figs. 7 and 8, respectively).

Further, comparison results on our own dataset, which contains samples collected from Google with various keywords, are shown in Figs. 9 and 10. As can be seen, the proposed method

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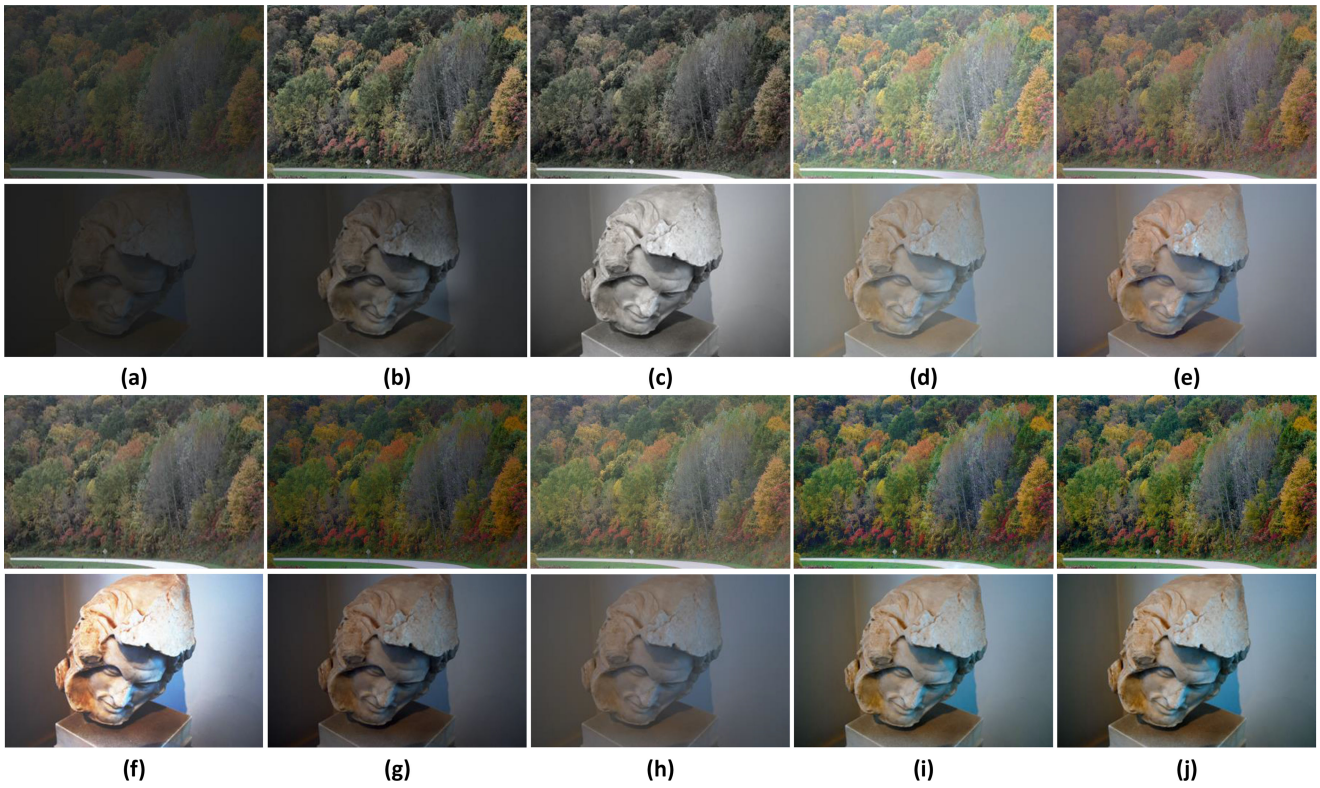


Fig. 7. Test samples from the MIT-Adobe FiveK dataset and corresponding enhancement results by previous methods and the proposed DSLR. (a) Original input. (b) CLAHE [22]. (c) LDR [25]. (d) LIME [27]. (e) HDRNet [11]. (f) DR [42]. (g) DPE [12]. (h) UPE [13]. (i) DSLR (proposed). (j) Ground truth.

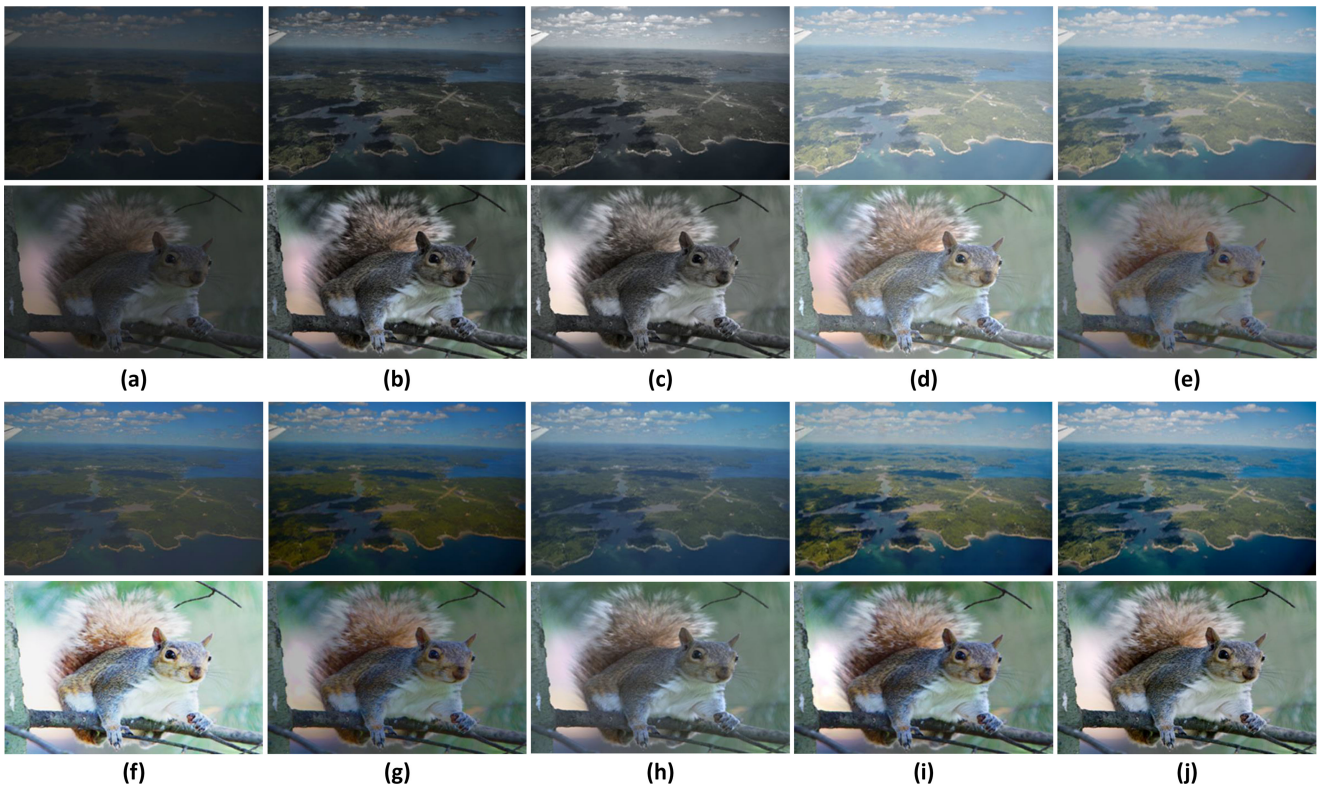


Fig. 8. More examples of enhancement results on the MIT-Adobe FiveK dataset. (a) Original input. (b) CLAHE [22]. (c) LDR [25]. (d) LIME [27]. (e) HDRNet [11]. (f) DR [42]. (g) DPE [12]. (h) UPE [13]. (i) DSLR (proposed). (j) Ground truth.

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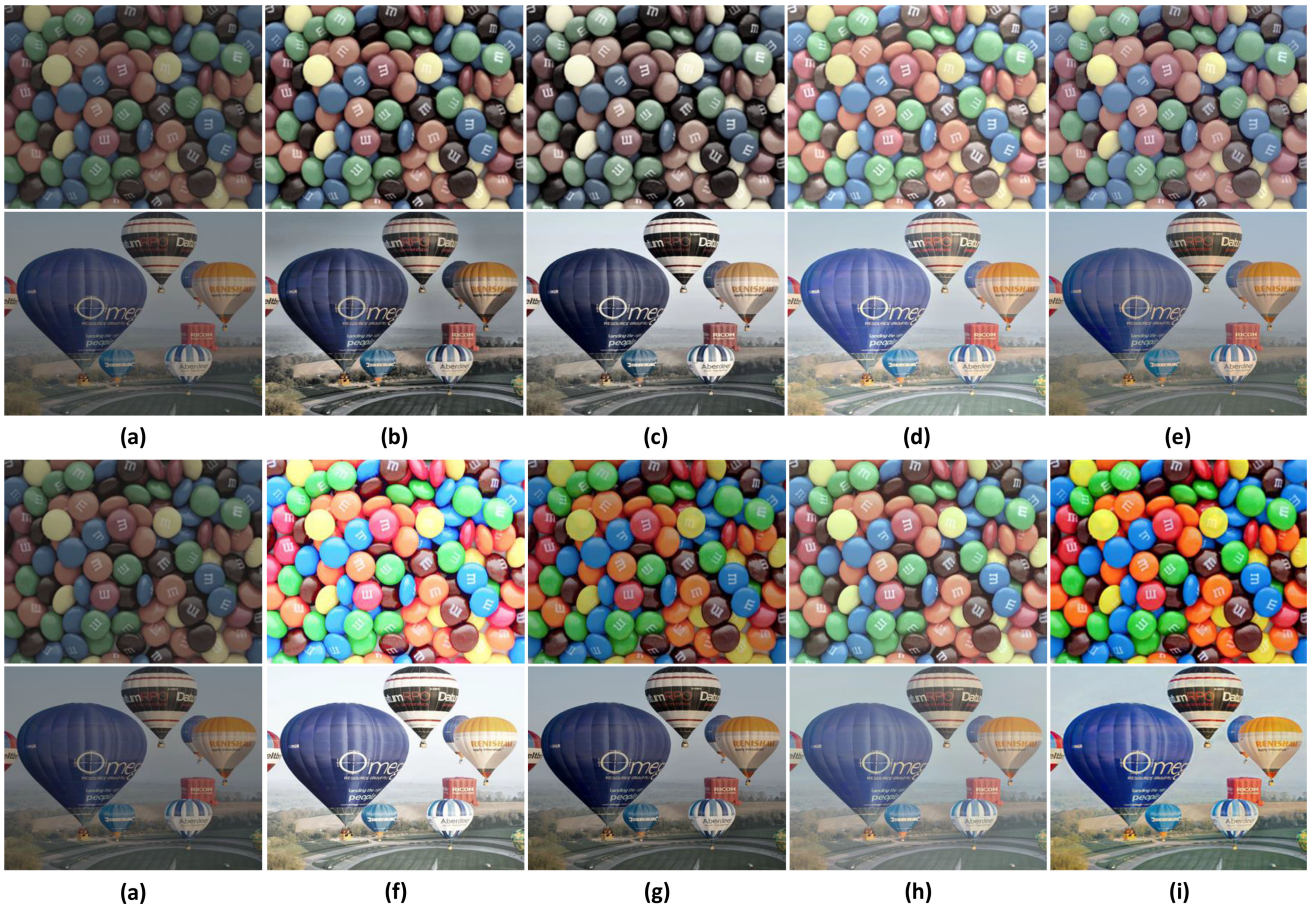


Fig. 9. Test samples from our own dataset and corresponding enhancement results by previous methods and the proposed DSLR. (a) Original input. (b) CLAHE [22]. (c) LDR [25]. (d) LIME [27]. (e) HDRNet [11]. (f) DR [42]. (g) DPE [12]. (h) UPE [13]. (i) DSLR (proposed).

moderately restores the contrast while efficiently preserving the naturalness, i.e., preventing the under- or over-enhancement, compared to previous methods (see the exteriors of chocolates and a building shown in Figs. 9 and 10, respectively). In contrast, CLAHE often generates undesirable artifacts at the boundary of the object, e.g., the ghost effect at the boundary of the hot-air balloon in Fig. 9(b). LDR, LIME, HDRNet, DR, and UPE have a limitation in moderately adjusting the contrast of enhancement results (i.e., under- or over-enhancement). Even though DPE shows reliable results for test samples, the overall brightness of the enhancement result is still dark compared to the proposed method. More examples of enhancement results by the proposed method on our dataset are shown in Fig. 11. From those examples, it is shown that the proposed method is able to generate visually acceptable results without under- or over-enhancement.

For the quantitative evaluation, we adopt six metrics, i.e., PSNR, SSIM [46], NIQMC [47], NIQE [48], BTMQI [49], and CaHDC [50], which have been widely employed for the performance comparison in the field of low-light image enhancement [13], [29]. Specifically, the first two metrics evaluate the performance according to the numerical distance and the structural similarity between the enhancement result and corresponding well-exposed image, i.e., ground truth, respectively. On the other hand, the last four metrics compute the score of the given image based on the entropy for informative areas, the quality-aware collection of the natural scene statistics (NSS) model, color tone mapping, and hierarchical perception mechanism (from local structure to global semantics) in human vision system in a no-reference manner, respectively. These visual quality metrics (VQMs) are expected to give the objective criteria to evaluate the performance of each method in diverse viewpoints. Note that the higher value of PSNR, SSIM, NIQMC, and CaHDC means the better performance for low-light image enhancement, and the other two metrics, i.e., NIQE and BTMQI, are the opposite.

TABLE II
PERFORMANCE COMPARISONS OF LOW-LIGHT IMAGE ENHANCEMENT BASED ON THE MIT-ADOBE FIVEK DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Methods	PSNR / SSIM	NIQMC / NIQE / BTMQI / CaHDC
Input	12.22 / 0.628	3.4929 / 5.6182 / 6.2784 / 29.1352
HDRNet [11]	22.04 / 0.855	5.0983 / 4.3297 / 4.6058 / 37.9481
DR [42]	20.89 / 0.815	5.1741 / 4.4800 / 4.4925 / 35.4050
DPE [12]	22.21 / 0.852	5.1494 / 4.2299 / 4.3672 / 38.0179
UPE [13]	23.10 / 0.878	5.2121 / 4.2244 / 4.3012 / 40.9512
Ours (DSLR)	24.34 / 0.904	5.2097 / 4.2099 / 4.2983 / 43.0649

Note: we compared ours with only learning-based methods since the MIT-Adobe FiveK dataset is constructed for learning the tone adjustment via ground truth.

Based on such metrics, we compared ours with eight previous methods on the MIT-Adobe FiveK dataset and our own dataset. The quantitative comparisons on each dataset are shown in Tables II and III, respectively. Note that the performance evaluation in our dataset is conducted with same parameters, which

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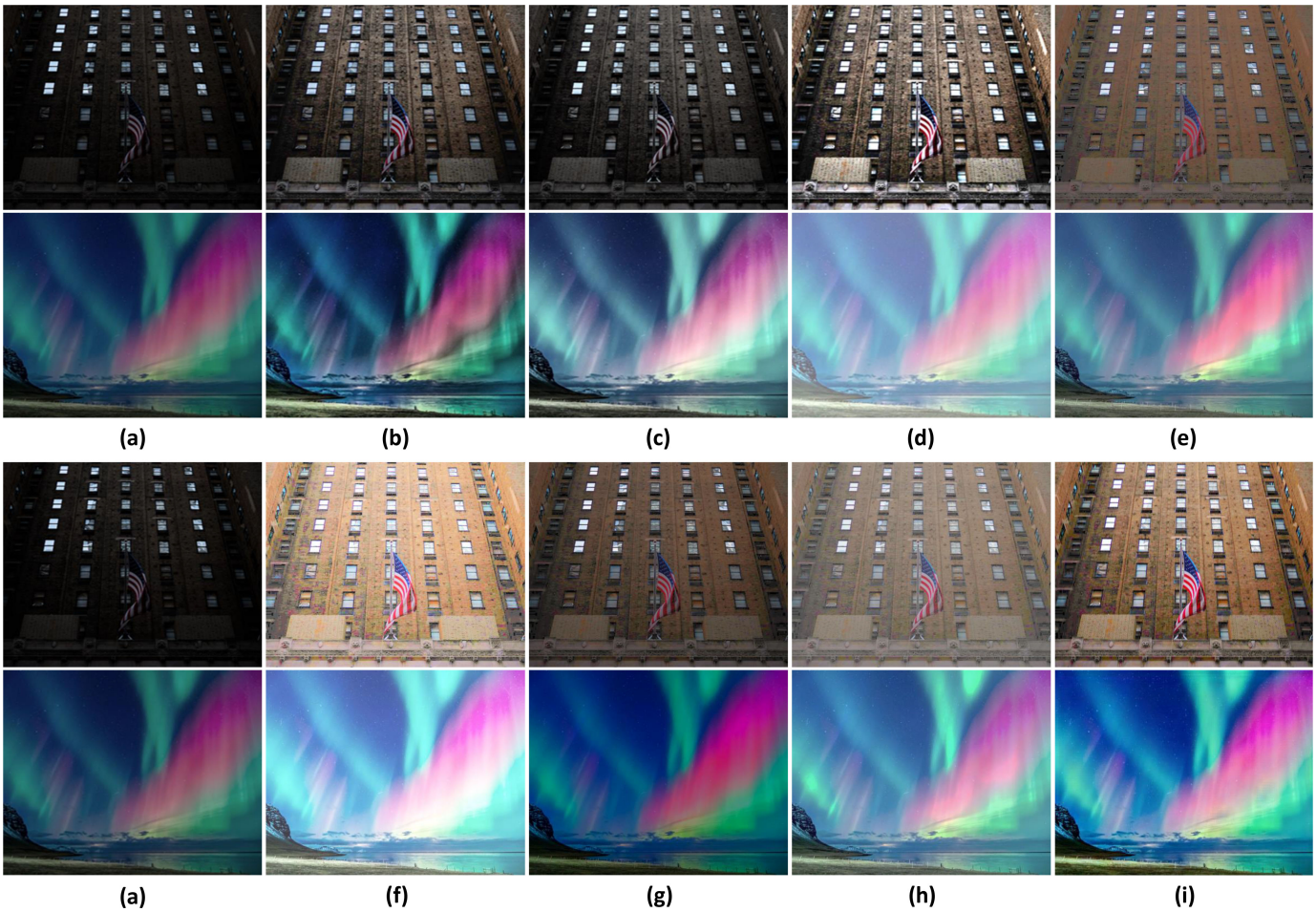


Fig. 10. More examples of enhancement results on our own dataset. (a) Original input. (b) CLAHE [22]. (c) LDR [25]. (d) LIME [27]. (e) HDRNet [11]. (f) DR [42]. (g) DPE [12]. (h) UPE [13]. (i) DSLR (proposed).



Fig. 11. More examples of enhancement results for our dataset. Top: original input images. Bottom: enhancement results by the proposed method.

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TABLE III
PERFORMANCE COMPARISONS OF LOW-LIGHT IMAGE ENHANCEMENT BASED ON OUR OWN DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Methods	Input	CLAHE [22]	LDR [25]	LIME [27]	HDRNet [11]	DR [42]	SICE [45]	DPE [12]	UPE [13]	Ours (DSLR)
NIQMC	4.1562	5.0251	4.9992	5.1203	5.0003	5.1130	5.1259	5.0925	5.0524	5.1310
NIQE	4.5108	3.8012	4.1094	3.8412	3.9048	3.9702	3.8942	3.7842	3.8911	3.7536
BTMQI	5.7067	4.4861	5.0923	4.2498	4.6008	4.4229	4.3009	4.6372	4.6087	4.2282
CaHDC	19.6713	36.4750	38.2726	38.2042	40.8347	39.4087	41.0873	39.7541	41.6703	41.7760

Note: our own dataset does not have ground truth, thus no-reference metrics (i.e., NIQMC, NIQE, BTMQI, and CaHDC) are only used for evaluating the performance of low-light image enhancement.

TABLE IV
COMPARISONS OF THE MODEL SIZE AND THE RUNNING TIME FOR THE RESOLUTION OF 1920 × 1080 PIXELS

Methods	CLAHE [22]	LDR [25]	LIME [27]	HDRNet [11]	DR [42]	SICE [45]	DPE [12]	UPE [13]	Ours (DSLR)
Model size	-	-	-	0.74M	33.40M	0.68M	2.20M	1.00M	14.31M
Running time	0.187s	0.674s	1.037s	0.021s	0.029s	0.049s	0.026s	0.036s	0.030s

are trained based on the MIT-Adobe FiveK dataset. As shown in Table II, the proposed method takes the first place in five quantitative metrics, i.e., PSNR, SSIM, NIQE, BTMQI, and CaHDC, and the second place in NIQMC. By adopting the Laplacian pyramid both in image and feature spaces, our DSLR is capable of effectively interpreting the embedding space, i.e., properties regarding the enhancement process, and thus achieves the performance improvement while providing visually acceptable results as shown in Figs. 7, 8, 9, 10 and 11. In particular, the performance in an aspect of PSNR is significantly improved over 1.24 dB compared with learning-based state-or-the-art methods. In order to show the generality of the proposed DSLR, we also evaluate the performance on our own dataset, which is constructed by using samples of various landscapes, as shown in Table III. In this dataset, the proposed DSLR outperforms eight representative methods in all of metrics. From those results, it is thought that our DSLR performs well under various lighting conditions. Even though the model size of the proposed method is relatively large compared to other learning-based methods as shown in Table IV, the proposed method still can be applied to real-time applications as like other methods without any additional processes (e.g., bilateral-grid guided upsampling), which are required to most previous methods for restoring the resolution of the result to the same as the original input.

C. User Study

To further show the objectivity of our enhancement results, we conducted a user study with 200 participants having various jobs, e.g., student, office worker, etc. To do this, the total 20 natural images are randomly chosen in our own dataset with corresponding enhancement results by previous methods and the proposed DSLR. Then, we utilized the Google platform for the survey and recruited participants through the online social media. Firstly, we showed 20 original inputs and corresponding enhancement results by previous methods and ours at the beginning of the survey, and then all enhancement results, which are generated by each method, were given so that participants could calculate the average score based on them without any interferes from other images. Participants evaluated the results of each method with a score between 1 and 5 according to their preference, i.e., the higher score means that the participant more prefers the corresponding enhancement result. The distribution of the preference rating is described in Fig. 12. As can be seen in this distribution, the proposed method is voted more than three-point by about 90 percent of people and takes the first place among eight image enhancers with a 3.77 score. From these experimental results, we could confirm that the proposed DSLR is quite efficient to resolve the problem of low-light image enhancement compared to previous methods in aspects of the user study as well as diverse quantitative evaluations.

TABLE V
PERFORMANCE ANALYSIS OF THE PROPOSED METHOD ACCORDING TO THE NUMBER OF LAPLACIAN PYRAMID LEVELS IN THE IMAGE SPACE (I.E., $K = 1$, 2, AND 3) AND THE TYPE OF THE RESIDUAL BLOCK (I.E., RESBLOCK [37] AND MSLB) ON THE MIT-ADOBE FIVEK DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Methods	PSNR / SSIM	NIQMC / NIQE / BTMQI / CaHDC
K = 1, w/ MSLB	21.68 / 0.820	5.0808 / 4.4309 / 4.7406 / 38.4192
K = 2, w/ MSLB	23.00 / 0.877	5.1853 / 4.2871 / 4.4021 / 41.8984
K = 3, w/ [37]	22.78 / 0.869	5.1489 / 4.3666 / 4.4872 / 40.3715
Proposed DSLR	24.34 / 0.904	5.2097 / 4.2099 / 4.2983 / 43.0649

Note: we did not investigate the model consisting of more than four pyramid levels in the image space due to the GPU memory limit.

D. Ablation Study

In this subsection, we try to confirm the effectiveness of network architectures, i.e., Laplacian pyramid both in image and feature spaces, and combinations of loss functions. To do this, we perform various comparative experiments and analyze their results. Firstly, performance variations are tested according to the number of Laplacian pyramid levels in the image space. As can be seen in the second and third rows of Table V, the models constructed with the relatively small number of Laplacian pyramid levels lead to the significant performance drop due to their lack of processes for enhancing local details. Note that the proposed DSLR (i.e., default model in Tables V and VI) adopts the three-level Laplacian pyramid in the image space, which is the highest pyramid level due to the GPU memory limitation, and is trained by using three loss functions. To verify the benefit of the proposed MSLB, we also test models based on our MSLBs and original ResBlocks (i.e., without the pyramid structure). As shown in Fig. 13 and Table V, the proposed MSLB

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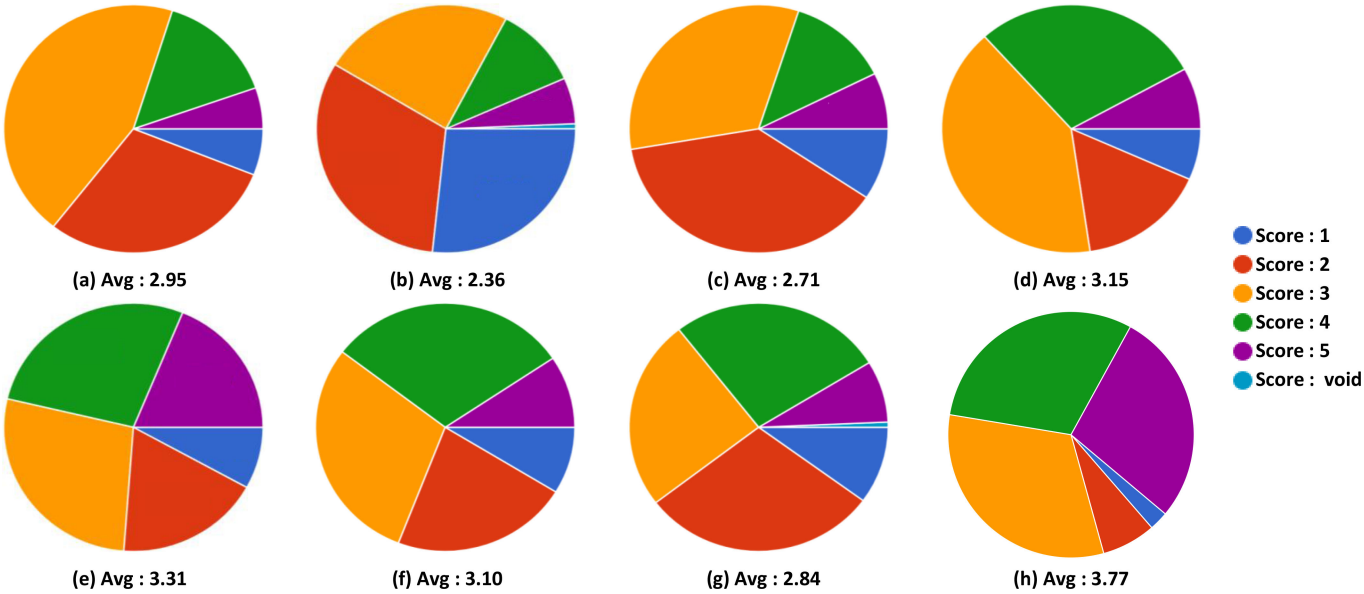


Fig. 12. Distributions of the preference rating for enhancement results by previous methods and the proposed DSLR on our own dataset. Each area, which is distinguished by six colors in the pie chart, means the rating frequency received from the participants. (a) CLAHE [22]. (b) LDR [25]. (c) LIME [27]. (d) HDRNet [11]. (e) DR [42]. (f) DPE [12]. (g) UPE [13]. (h) DSLR (proposed). Note that the "void" indicates the case in which the participant gave up on giving the score for the enhancement result.

TABLE VI
PERFORMANCE ANALYSIS OF THE PROPOSED METHOD ACCORDING TO COMBINATIONS OF LOSS FUNCTIONS ON THE MIT-ADOBE FIVEK DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Methods	PSNR / SSIM	NIQMC / NIQE / BTMQI / CaHDC
w/L_d, w/oL_1,L_c	22.90 / 0.870	5.1434 / 4.3382 / 4.5507 / 40.4172
w/L_d,L_1, w/oL_c	23.78 / 0.885	5.1869 / 4.2900 / 4.3653 / 42.0744
Proposed DSLR	24.34 / 0.904	5.2097 / 4.2099 / 4.2983 / 43.0649

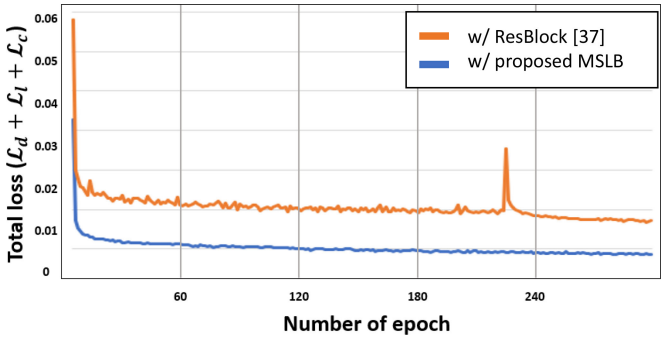


Fig. 13. Comparison of the training accuracy according to the type of the residual block.

TABLE VII
PERFORMANCE ANALYSIS OF THE PROPOSED METHOD ACCORDING TO DIFFERENT NUMBERS OF MSLBS IN EACH RESTORER ON THE MIT-ADOBE FIVEK DATASET (THE BEST RESULTS ARE SHOWN IN BOLD)

Methods	PSNR / SSIM	NIQMC / NIQE / BTMQI / CaHDC
w/ one MSLB	20.95 / 0.776	5.1082 / 4.2950 / 4.3409 / 38.2462
w/ two MSLBs	22.44 / 0.851	5.1571 / 4.2416 / 4.3274 / 41.3218
w/ three MSLBs	24.34 / 0.904	5.2097 / 4.2099 / 4.2983 / 43.0649
w/ four MSLBs	24.38 / 0.909	5.1894 / 4.1970 / 4.3009 / 42.9974
w/ five MSLBs	24.46 / 0.916	5.1930 / 4.1961 / 4.2960 / 43.0301

Note: each restorer in the default setting of the proposed method consists of three MSLBs.

plays an important role to guarantee the stability while showing the relatively small loss, which brings the performance progress of the proposed method, during the training phase. Note that the two-level and three-level MSLB are replaced with two and three ResBlocks for constructing the original ResBlock-based model, respectively. Further, various combinations of loss functions are tested quantitatively as shown in Table VI. Based on this result, we observe the advantage of each loss for enhancing the performance in five metrics and accurately restoring the degraded image as shown in Fig. 6. The best performance is obtained when three loss functions, i.e., data loss $\mathcal{L}_d$, Laplacian loss $\mathcal{L}_l$, and color loss $\mathcal{L}_c$, are used for training our DSLR. In Table VII, the performance of the proposed method is improved by using multiple MSLBs. However, the performance becomes almost saturated in cases of more than three MSLBs while computational costs and the training time are increased. To balance in three aspects, i.e., performance, computation costs, and training time, we choose three MSLBs in each restorer as the default setting. Consequentially, experimental results show that the proposed DSLR paves the way for low-light image enhancement by providing reliable restoration results based on the decomposition-based scheme.

E. Limitation

For our future research, the limitations of the proposed method are discussed in this subsection. Some failure cases of the proposed method are shown in Fig. 14. As can be seen in these examples, the proposed method often fails to restore images, which are degraded by extremely low-light conditions, thus the result still appears to be underexposed. Furthermore, the problem of noise amplification also occurs when images acquired

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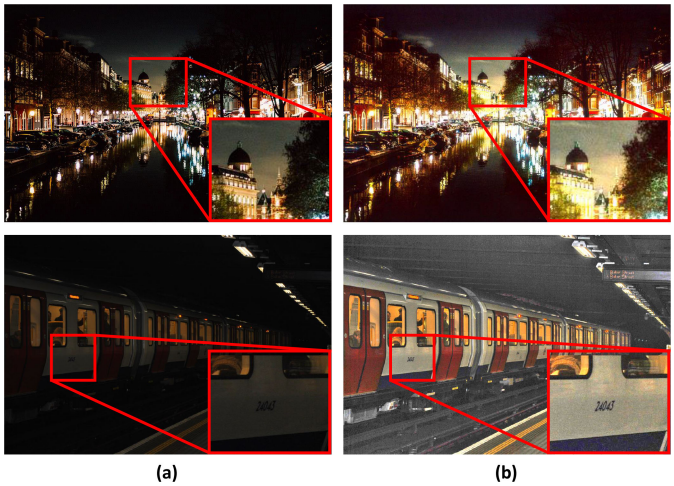


Fig. 14. Examples of noise amplification in enhancement results. (a) Input images. (b) Enhancement results by the proposed method. Note that the noisy region is enlarged.

in such environments are given for the test. To cope with these limitations, our future work is to implicitly incorporate the denoising process into the proposed DSLR and to strengthen the ability of the proposed method for providing visually acceptable results even under extremely low-light conditions.

V. Conclusion

In this paper, a novel method for low-light image enhancement has been proposed. By adopting the Laplacian pyramid in image and feature spaces, which is the key idea of the proposed method, the complicated problem of low-light image enhancement can be resolved as separately enhancing the global illumination and local details in different streams of the encoder-decoder architecture. Specifically, each decomposed image is adjusted by the corresponding restorer and the output is progressively combined to generate the final enhancement result. Moreover, the proposed multiscale Laplacian-residual block (MSLB) gives a great help to make optimization easier by improving the flow of information via abundant connections of higher-order residuals in a multiscale manner, which lead to the relatively small loss with a stable convergence in the training phase. Experimental results on various datasets show that the proposed method outperforms state-of-the-art approaches and can be applied to restore various degraded images acquired from complicated lighting environments.

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Seokjae Lim received the B.S. degree from the Department of Electronics Engineering, Konkuk University, Seoul, South Korea. He is currently working toward the graduate degree with the Department of Electronic, Information, and Communication Engineering, Konkuk University, Seoul, South Korea. His research interests include image enhancement, computer vision, deep learning, and biometrics.



Wonjun Kim received the B.S. degree with the Department of Electronic Engineering, Sogang University, Seoul, South Korea, the M.S. degree from the Department of Information and Communications, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, and the Ph.D. degree from the Department of Electrical Engineering, KAIST, in 2006, 2008, and 2012, respectively. From September 2012 to February 2016, he was a Research Staff Member with the Samsung Advanced Institute of Technology (SAIT), Gyeonggi-do, South Korea. Since March 2016, he has been with the Department of Electrical and Electronics Engineering, Konkuk University, Seoul, South Korea, where he is currently an Associate Professor. His research interests include image and video understanding, computer vision, pattern recognition, and biometrics, with an emphasis on background subtraction, saliency detection, face and action recognition. He has served as a regular reviewer for more than 30 international journal papers, including the IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE ACCESS, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON CYBERNETICS, IEEE SIGNAL PROCESSING LETTERS, PATTERN RECOGNITION, etc.

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